

QuietPrune: Query-Guided Early Token Pruning for Vision-Language Models

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Abstract

Vision-language models (VLMs) demonstrate powerful capabilities in multimodal tasks. However, the large number of visual tokens imposes a significant computational cost. In this paper, we propose *QuietPrune*, a **Q**Uery-gu**I**ded **E**arly **T**oken **P**run**I**ng method to remove redundant visual tokens within VLMs, thereby enhancing computational efficiency. Unlike previous late pruning methods, we recognize that implementing early pruning within the vision transformer (ViT) can achieve benefits in both latency reduction and accuracy maintenance. To address the semantic loss problem in early pruning, we design a lightweight adapter by performing an inverse transformation of the projector in VLMs. The proposed adapter converts the contextual query into a visual domain [Q-CLS] (Query [CLS]) token, providing textual guidance for ViT pruning. During pruning, we further introduce a semi-structured pruning scheme based on visual-textual relevance. Specifically, we group spatially adjacent 2×2 tokens to accommodate the visual token merging operation prevalent in mainstream VLMs. We use the mean attention scores between the [Q-CLS] token and the visual tokens as the relevance metric for each group, avoiding additional computation. Pruning is then applied at the group level based on the relevance score, preserving positional continuity. After pruning, we aggregate the redundant tokens into a single token to maintain context cues. Our method achieves up to 19.0% reduction in prefill latency while outperforming 4.2% in accuracy on the recent Qwen3-VL and InternVL3 series compared to existing late pruning methods.

1. Introduction

Vision-language models (VLMs) [1, 2, 27, 37, 40, 47] demonstrate revolutionary performance on complex multimodal tasks. State-of-the-art VLMs typically comprise three components: a vision transformer (ViT), a projector, and a large language model (LLM). Despite its power, this architecture faces a computational efficiency challenge. The number of visual tokens produced by ViT scales quadrati-

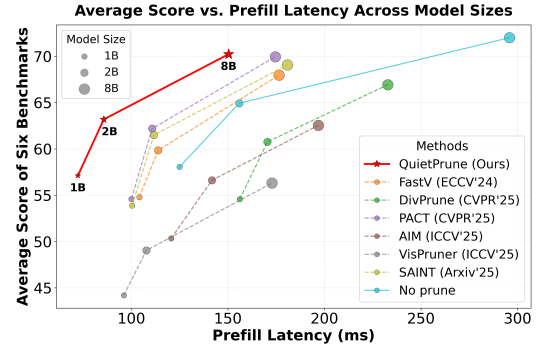


Figure 1. Comparison with SOTA token pruning methods on the InternVL3 series [47]. The proposed QuietPrune method outperforms the existing methods on both accuracy and prefill latency.

cally with the input resolution, making VLM deployment prohibitively expensive for real-world applications.

To eliminate computational bottlenecks, researchers propose token pruning techniques [3, 10, 14, 20, 36, 44, 46] that aim to remove redundant visual tokens in real-time. However, existing methods adopt a late pruning strategy. Specifically, late pruning refers to the strategy of removing tokens after the ViT has completed its full computation on all visual tokens, typically at the interface between the ViT and the LLM or within the LLM layers. While these approaches provide efficiency improvements, they overlook the inherent computational redundancy within the token generation stage of ViT, especially on small-sized VLMs. Moreover, the computational cost associated with the pruning decision mechanism is commonly overlooked in these methods. For some methods, the time required for token selection can even surpass the time saved by token pruning, leading to a paradox where pruning actually degrades prefill speed.

Compared to late pruning, pruning inside the ViT (which we refer to as early pruning) offers the advantage of reducing computational cost during the ViT stage. To quantify the potential of early pruning, we profile the prefill latency of Qwen3-VL [1] and InternVL3 [47] across multiple model scales and input configurations. InternVL3 adaptively partitions high-resolution images into multiple tiles

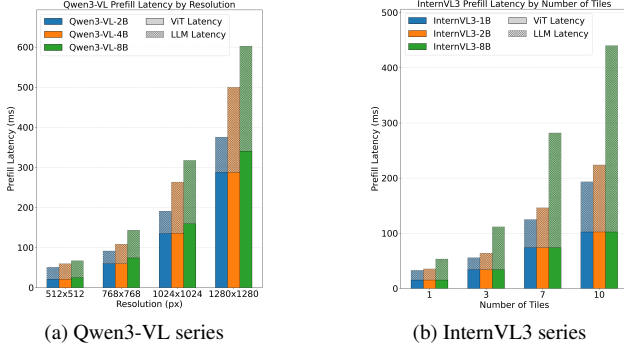


Figure 2. Latency distribution between ViT and LLM during the prefill phase for the Qwen3-VL and InternVL3 model series across different model sizes with various input resolutions or number of tiles. We report the latency on a single A100 GPU. The ViT accounts for more than 50% of the latency in most settings.

(sub-images) for processing. As shown in Fig. 2, ViT computation dominates the prefill phase, accounting for over 50% of latency across most settings and exceeding 75% for smaller models with high-resolution inputs. This distribution reveals that early pruning within ViT can yield substantially greater efficiency gains than late pruning strategies that only reduce LLM computation.

However, early pruning in VLMs presents unique challenges. Conventional ViT pruning methods [12, 25, 34] rely solely on visual saliency to identify “important” patches. Early pruning occurs before text-vision interaction, lacking guidance on which regions are query-relevant. Consequently, visually salient tokens (e.g., foreground objects) may be retained while query-critical tokens (e.g., small text) are discarded. As illustrated in Fig. 3, given “What number is the building?”, conventional methods erroneously remove tokens encoding the building number, which is visually inconspicuous but semantically critical. This leads to irrecoverable information loss. Unlike late pruning, where tokens have passed through text-conditioned layers, early pruning needs textual relevance without cross-modal signals.

Furthermore, the latest mainstream VLMs usually employ a visual token merging operation [5, 13, 47] to reduce the number of visual tokens fed into the LLM. Commonly, spatially adjacent 2×2 patches are merged into a single visual token, resulting in a reduction of the count of visual tokens by $4\times$. Preserving spatial contextual information, which is overlooked in conventional unstructured pruning, poses another challenge in early pruning.

To address the aforementioned challenges, we propose a **QUery-guided Early Token Pruning** (QuietPrune) method for vision-language models. Figure 4 illustrates the overall framework of our method. To address the semantic misalignment in early pruning, we design a lightweight text-to-vision adapter that converts a textual query into ViT-

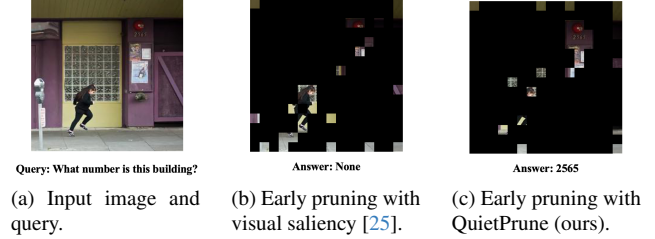


Figure 3. Comparison of pruning results with different methods. With the query-guided scheme, our method retains tokens relevant to the input query and makes an accurate prediction.

compatible visual domain features. These features are pooled into a [Q-CLS] (Query [CLS]) token to provide textual guidance for ViT pruning. Crucially, we perform an inverse transformation approach to initialize the parameters of the adapter from the pretrained vision-to-text projector in VLMs. This approach enables the proposed adapter to effectively convert text embeddings into the visual domain with minimal training. We further propose a semi-structured pruning strategy based on visual-textual relevance. This strategy takes the attention scores between the [Q-CLS] token and visual tokens as a relevance measure. These scores are naturally produced by ViT during inference without incurring extra latency overhead. To accommodate the token merging operation in mainstream VLMs, our semi-structured pruning groups spatially adjacent 2×2 visual tokens and performs group-wise pruning based on their average relevance scores. This strategy avoids positional distortion associated with random pruning and maintains compatibility across a wide range of VLM architectures. After pruning, we aggregate the redundant tokens into a single token to capture useful context cues. During training, we update only the parameters of the adapter while keeping the VLM frozen. We employ a self-distillation scheme for adapter training. Based on the above design, our method outperforms existing methods on multiple VLMs, reducing latency by up to 19% while improving accuracy by 4.2%. Our contributions can be summarized as follows:

- We propose the first query-guided early pruning scheme for VLMs, where textual queries guide token pruning during ViT’s forward pass rather than after feature extraction.
- We propose a lightweight text-to-vision adapter via inverse transformation of the pretrained projector, enabling textual guidance in early ViT pruning with minimal training and negligible computational cost.
- We introduce a semi-structured pruning strategy on 2×2 token groups, preserving positional embedding stability and ensuring compatibility with mainstream visual token merging operations in modern VLMs.
- Extensive experiments show that our method outperforms existing methods on multiple VLM architectures.

2. Related work

2.1. Vision-language model

The breakthrough success of large language models (LLMs) [2, 4, 39] in textual understanding [18, 45] and generation [11] has catalyzed extensions to multimodal reasoning. This has spurred the development of vision-language models (VLMs) [2, 27, 37, 40, 47] that bridge visual and textual domains. The typical VLM architecture comprises three key components: a vision transformer (ViT), a projector, and an LLM. Popularized by Blip-2 [24], this paradigm has become standard in frameworks like LLaVA [27], which combines CLIP [33] with Vicuna [38] to achieve strong multi-modality reasoning capabilities. While recent works aim to enhance fine-grained visual understanding by scaling input resolutions [28, 29], this leads to a quadratic growth in visual tokens, introducing substantial computational overhead.

To reduce computational cost on the LLM, methods like InternVL [47] and Qwen-VL [40] employ a token merging operation on visual tokens. InternVL merges spatially adjacent 2×2 patches via a pixel shuffle solution, while Qwen-VL utilizes an MLP-based merging strategy. Both approaches achieve a $4 \times$ reduction in visual tokens before LLM processing. However, these methods do not address the dominant computational burden in the ViT.

2.2. Visual token reduction

Visual token reduction techniques improve model efficiency by dynamically removing redundant visual tokens. These methods fall into two categories: token pruning [3, 10, 16, 41] and token merging [7, 35, 42]. Token pruning directly removes redundant tokens, while token merging combines the information of redundant tokens into important tokens. Since token merging typically involves additional token matching and fusion operations that incur extra latency, this work focuses on token pruning techniques.

In general vision tasks, researchers leverage visual saliency for token pruning. EViT [25] prunes tokens using attention scores relative to the [CLS] token. DynamicViT [34] employs an auxiliary predictor to predict the redundant tokens. Though effective for pure vision tasks, these methods face adaptation challenges in multimodal scenarios.

Token pruning schemes for VLM have rapidly emerged in recent years. FastV [10] evaluates token importance via average attention scores of visual tokens. VTW [26] initiates visual token pruning in deeper layers of the LLM and completely removes visual tokens in the final layers. Recent studies focus more on improving pruning metrics. DivPrune [3] formalizes token pruning as an MMDP problem to maximize the diversity of retained tokens. PACT [14] adopts similarities between visual token keys and a global

query as importance metrics. However, all these methods perform pruning either at the LLM input or within its internal layers, overlooking the redundancy inherent in the ViT.

The recent study SAINT [20] introduces early ViT pruning via similarity-based bipartite matching, but it uses only visual saliency as guidance and ignores textual cues, thereby suffering substantial performance drops in multimodal settings.

3. Preliminaries

3.1. Vision-language Models

Mainstream VLMs typically consist of three components: a ViT, a projector, and an LLM. For a given image-text pair (X_q, X_v) , the ViT first converts the image input X_v into visual features F_v . The projector maps F_v into the textual domain to obtain visual tokens H_v . The LLM then predicts the answer from H_v and the text embedding H_q (obtained by embedding the text input X_q) in an auto-regressive manner. The entire pipeline can be formulated as:

$$F_v = g(X_v) \quad (1)$$

$$H_v = \text{proj}(F_v), H_q = \text{emb}(X_q) \quad (2)$$

$$p(y_t) = \prod_{t=1}^N p_{\theta}(y_t | H_v, H_q, y < t), \quad (3)$$

where $g(\cdot)$, $\text{proj}(\cdot)$, and $\text{emb}(\cdot)$ are the functions of ViT, projector, and embedding, respectively. y_t is t -th prediction during auto-regression. p is the conditional probability. θ is the parameter of LLM.

3.2. Visual token pruning

Despite the strong capabilities of VLMs, they face severe challenges in computational efficiency. An effective acceleration approach is visual token pruning, which removes redundant visual tokens during VLM inference. The problem of visual token pruning can be formulated as:

$$H_v^* = f(H_v) \quad (4)$$

$$p^*(y_t) = \prod_{t=1}^N p_{\theta}(y_t | H_v^*, H_q, y < t) \quad (5)$$

$$\text{Objective} : \min_f \mathcal{L}(p^*(y_t), p(y_t)), \quad (6)$$

where H_v^* are the retained tokens after pruning, f is the pruning operation and $\mathcal{L}$ is the loss function that measures the difference between the model's outputs with and without pruning.

4. Method

We propose QuietPrune, a query-guided early pruning framework that adaptively removes redundant visual tokens

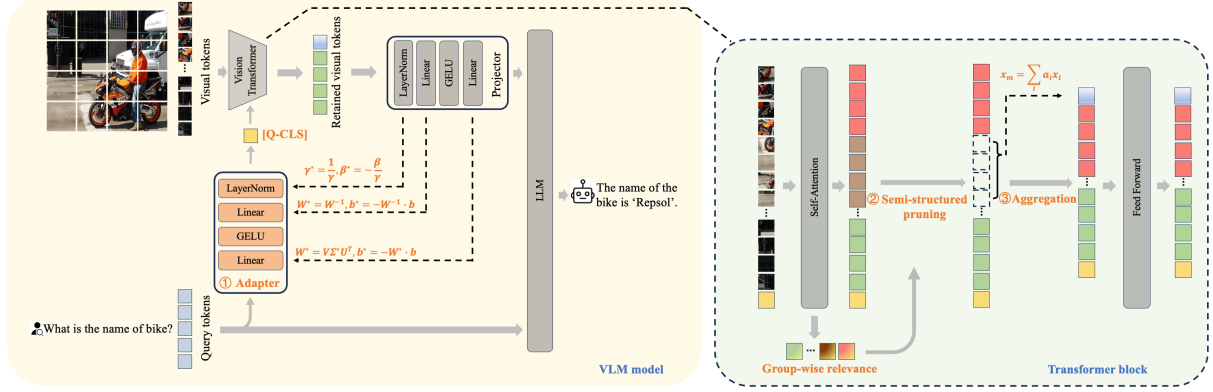


Figure 4. Overall framework of QuietPrune. We perform a query-guided early pruning strategy. An adapter is proposed to convert text embeddings into the visual domain. The adapter leverages the capability of the projector through an inverse transformation and generates a [Q-CLS] token to guide the pruning process of ViT. We propose a semi-structured pruning based on visual-textual relevance. The visual tokens are pruned in a group-wise manner to preserve the positional information. After pruning, we aggregate the redundant tokens into a single token to maintain context cues.

within the ViT. As illustrated in Fig. 4, our method comprises three components: (1) a lightweight text-to-vision adapter that projects query tokens into visual space for query-guided pruning; (2) a semi-structured pruning mechanism that removes low-relevance token groups while preserving spatial coherence; and (3) a token aggregation module that consolidates pruned tokens to retain complementary information. The proposed adapter is initialized via inverse transformation of the existing vision-to-text projector and optimized through self-distillation training, while keeping all parameters of the VLM fixed.

4.1. Text-to-Vision adapter via projector inversion

To enable query-guided early pruning, we introduce a lightweight adapter that produces textual information for the ViT encoder. In typical VLMs, the vision-to-text projector maps visual tokens to the textual embedding space using a stack of LayerNorm, GELU, and two linear layers. As shown in Fig. 4, we reverse this architecture to construct a text-to-vision adapter by applying the same components in opposite order. The proposed adapter converts query embeddings back into the visual feature space. The resulting features are pooled into a single [Q-CLS] token, which is then prepended to the visual token sequence and processed jointly within ViT to guide pruning decisions.

For adapter initialization, we obtain the weight and bias by applying the inverse transformation to the existing projector in the VLM. This ensures that the adapter has a fundamental capability to map query tokens to the visual domain. Specifically, for a linear layer in the projector with the forward pass:

$$Y = WX + b, \quad (7)$$

where X and Y denote the input and output of the linear

layer, respectively, and W and b are its weight and bias. When W is invertible, we initialize the weight W^* and bias b^* of the corresponding linear layer in the adapter with the inverse parameters:

$$W^* = W^{-1}, b^* = -W^{-1} \cdot b. \quad (8)$$

For non-invertible weight matrix W , we employ the Moore-Penrose pseudoinverse [6] to minimize the reconstruction error $\|X - W^*Y\|$:

$$W^* = \lim_{\alpha \rightarrow 0^+} (W^T W + \alpha I)^{-1} W^T. \quad (9)$$

This admits a closed-form solution via singular value decomposition (SVD). Let

$$W = U \Sigma V^T, \quad (10)$$

where U and V are orthogonal matrices of left and right singular vectors, and Σ is the diagonal matrix of singular values. The pseudoinverse is then given by:

$$W^* = V \Sigma^* U^T, \quad (11)$$

where Σ^* is obtained by transposing the reciprocal of non-zero elements in Σ .

For LayerNorm and GELU, exact inverse transformations do not exist. To approximately invert the LayerNorm operation in the projector, we assume its output follows a standard normal distribution. The scale and bias parameters (γ^*, β^*) of the LayerNorm in the adapter can be initialized with:

$$\gamma^* = \frac{1}{\gamma}, \beta^* = -\frac{\beta}{\gamma}, \quad (12)$$

where γ and β are the scale and bias parameters of the LayerNorm in the projector, respectively.

For GELU, we retain the same activation in the adapter because it closely approximates the identity function for positive inputs. This helps preserve signal fidelity in the linear regime of the activation.

With the inverse transformation strategy described above, the adapter requires only minimal training on a small dataset to achieve effective text-to-vision feature transformation.

4.2. Semi-structured pruning based on visual-textual relevance

The [Q-CLS] token, generated by the adapter, is prepended to the sequence of visual tokens to form the full input $X \in \mathbb{R}^{(N+1) \times d}$ for the ViT. Within each attention layer, queries Q , keys K , values V and output O are computed as:

$$Q = XW_Q, K = XW_K, V = XW_V \quad (13)$$

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right), \quad (14)$$

$$O = AV, \quad (15)$$

where N denotes the number of visual tokens, d is the embedding dimension and A is the attention scores. The first row of O corresponds to the updated [Q-CLS] token, which fuses textual semantics with global visual context through attention.

The attention scores in A naturally reflect the relevance between token pairs. Therefore, we use the attention scores between the [Q-CLS] token and all visual tokens as a measure of each visual token’s relevance to the input query. Since these scores are naturally computed during ViT inference, leveraging them as relevance indicators introduces no additional latency overhead.

In current mainstream VLMs, visual tokens are spatially merged before being passed to the language model. As shown in Fig. 5, adjacent 2×2 tokens are merged into a single token for the LLM, reducing the number of visual tokens by $4\times$. Under this scheme, unstructured pruning can disrupt spatial coherence: after arbitrary token removal, the remaining elements within a merging group may no longer be spatially contiguous. This fragmentation breaks the local structural prior assumed by the merging operation and can introduce significant prediction bias.

To preserve spatial integrity, we perform semi-structured pruning, where each 2×2 block of adjacent tokens is treated as a group. The visual-textual relevance of a group is defined as the average of the relevance scores of its constituent tokens. During pruning, groups with higher relevance scores are retained in full, while low-relevance groups are removed as indivisible units. By preserving the spatial consistency of the retained tokens, this design ensures our method is compatible with diverse VLM architectures.

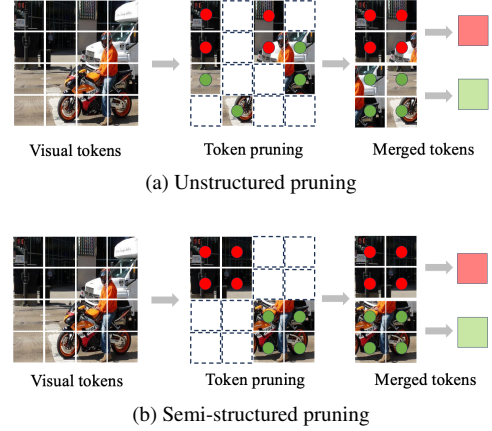


Figure 5. Comparison of unstructured pruning and semi-structured pruning. Unstructured pruning can distort patch positional information, whereas our semi-structured pruning method can preserve spatial continuity.

4.3. Redundant token aggregation

Through our semi-structured pruning strategy, redundant visual tokens are removed within the ViT. However, discarding these tokens entirely may lead to loss of useful contextual cues. Therefore, we aggregate the pruned tokens into a single compact representation. Specifically, the aggregated token x_m is computed as a weighted sum of the pruned tokens x_i , with each token weighted by its relevance score a_i as $x_m = \sum_i a_i x_i$. The aggregated token is then concatenated with the retained visual tokens and passed to the language model for prediction. Since only one aggregated token is added, the latency overhead is negligible.

4.4. Self-distillation training

During training, we update only the parameters of the proposed adapter while keeping the VLM frozen. According to Eq. (6), we minimize the output discrepancy between the pruned and un-pruned models by treating the pruned model as the student and the un-pruned model as the teacher in a self-distillation framework. The distillation loss $L_{distill}$ is the KL divergence between teacher logit Y_t and student logit Y_s . The total loss L_{total} is combined with the distillation loss and cross-entropy loss L_{ce} between the logits of the student and the ground-truth target Y_{gt} , which is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{distill}(Y_s, Y_t) + \mathcal{L}_{ce}(Y_s, Y_{gt}) \quad (16)$$

5. Experiments

5.1. Experimental settings

Benchmark: We evaluate QuietPrune on a diverse set of vision-language benchmarks based on VLMEvalKit [15], covering multiple dimensions of VLM performance.

Specifically, we use MMBench [30] and MMStar [9] for comprehensive multimodal evaluation, MMMU [43] for visual reasoning, AI2D [22] and OCRBench [31] for text recognition and comprehension, and HallusionBench [17] to assess hallucination robustness.

Models: We conduct experiments on recent vision-language models, including the Qwen3-VL [1] and InternVL3 [47] series. To assess generalization across model scales, we test multiple variants: InternVL3-1B, Qwen3-VL-4B, and InternVL3-8B. Additional experiment results on more models are provided in the supplementary materials.

Implementation details: The adapter in our method is trained on a mixture of publicly available open-source datasets, including LLaVA-Instruct [27], ShareGPT4V [8], ChartQA [32], ScreenQA [19], SynthDoG [23], and MultiVQA [21]. We randomly sample 10K data in total (0.8% of the combined dataset), which requires approximately 40 minutes of training on a single NVIDIA A100 GPU. We apply token pruning at three fixed layers of the ViT, namely at depths $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{3}{4}$ of the total number of blocks. At each of these layers, a fraction R of visual tokens is pruned based on group relevance scores. During training, R is set to 25%. During evaluation, R can be adjusted flexibly without retraining.

Evaluation setup: The original pruning configurations are different across methods. For a fair comparison, we evaluate all methods under the same average visual token pruning rate at the LLM stage. For an LLM with L transformer layers, the average pruning rate R^* is defined as: $R^* = 1 - \frac{\sum_{i=1}^L \hat{T}_i}{\sum_{i=1}^L T_i}$, where T_i and $\hat{T}_i$ are the number of visual tokens in layer i with and without pruning, respectively. The specific pruning configurations for each method under different average pruning rates are provided in the supplementary materials.

We report two key metrics across the six benchmarks: relative accuracy and prefill latency reduction. Specifically, relative accuracy is defined as $RA = \frac{acc_p}{acc_{np}}$, where acc_p and acc_{np} denote the accuracies of the pruned and un-pruned models, respectively. Latency reduction is computed as $LR = \frac{lat_{np} - lat_p}{lat_{np}}$, where lat_p and lat_{np} are the prefill latencies of the model with or without pruning. All latency measurements are obtained on a single NVIDIA A100 GPU and averaged across all samples in each benchmark. Following [20], to ensure consistent inference conditions, we use standard eager-mode attention for all methods.

5.2. Comparison with SOTA methods

We compare QuietPrune against the SOTA VLM pruning methods on the Qwen3-VL [1] and InternVL3 [47] series. The methods include PACT [14], SAINT [20], DivPrune [3], AIM [46], FastV [10], and VisPruner [44]. For

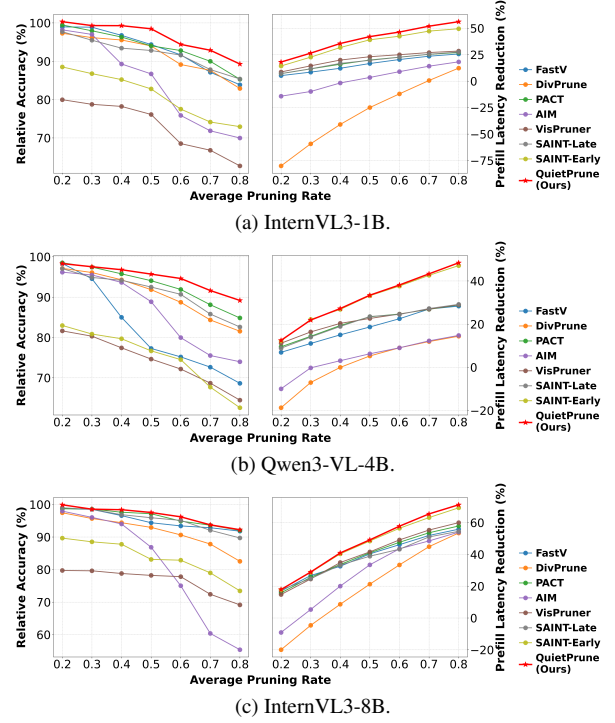


Figure 6. Comparison of different methods on the InternVL3 and Qwen3-VL series. Left is the relative accuracy, and right is the latency reduction of different methods under various pruning rates.

SAINT, we report both its early-pruning and late-pruning variants.

As shown in Fig. 6, our method consistently outperforms the other methods in both relative accuracy and latency reduction across different VLM families and model sizes. Compared to SAINT’s early pruning variant (SAINT-early), QuietPrune maintains over 88% accuracy at an 80% pruning rate, whereas SAINT’s accuracy drops below 90% even at a 20% pruning rate. This gap arises because SAINT relies on unstructured, visual-saliency-based token pruning, which disrupts semantic and spatial coherence among retained tokens. In contrast, our query-guided, semi-structured pruning method selectively preserves the textually relevant tokens while maintaining their spatial consistency.

Compared to late pruning methods, our method achieves significant latency reduction. This advantage becomes more pronounced as model size decreases, because ViT accounts for a larger fraction of the total computation in smaller VLMs, making them more amenable to early pruning. Notably, the pruning procedures of DivPrune and AIM are computationally expensive, leading to total prefill latency that exceeds the un-pruned baseline.

In terms of accuracy, AIM suffers significant performance drops beyond 40% pruning, indicating that full token removal is only effective in deep layers. FastV shows poor

Table 1. Main results on the InternVL3 and Qwen3-VL series. All methods achieve an average pruning rate of 50% in the LLM part. “SAINT-Early” and “SAINT-Late” represent the SAINT method with early pruning and late pruning, respectively. “acc” refers to the accuracy score of the benchmark. “lat” refers to the prefill latency (ms) of each method. “RA%” refers to the relative accuracy compared to the model without pruning. “LR%” refers to the latency reduction compared to the model without pruning. Negative values in “LR%” indicate that the latency of the pruned model is larger than that of the model without pruning. We use **red** to highlight the best results of all methods and **blue** to highlight the best results of late pruning methods. Our method outperforms other methods on different model sizes and architectures.

Model	Method	MMBench acc lat	MMStar acc lat	MMMU acc lat	Hallusion acc lat	AI2D acc lat	OCRBench acc lat	Avg acc lat	RA% LR%
InternVL3-1B	No prune	69.1 85.2	52.3 88.7	41.3 174	36.3 88.8	69.8 83.3	79.7 230	58.1 125	- -
	FastV (ECCV’24)	67.3 77.9	50.6 80.1	41.2 137	36.1 80.4	66.7 75.2	66.9 174	54.8 104	94.3 16.8
	DivPrune (CVPR’25)	67.2 110	51.2 111	41.8 217	36.6 112	67.1 105	63.6 283	54.6 156	94.0 -24.8
	PACT (CVPR’25)	67.2 75.2	50.1 77.3	40.2 132	36.7 77.2	67.1 71.8	66.2 167	54.6 100	94.0 20.0
	AIM (ICCV’25)	68.0 99.0	52.4 101	40.4 145	37.3 101	69.3 94.6	34.6 184	50.3 121	86.6 3.20
	VisPruner (ICCV’25)	62.7 72.6	46.1 74.2	38.8 126	33.4 74.5	58.8 69.2	25.4 160	44.2 96.1	76.1 23.1
	SAINT-Late (Arxiv’25)	66.7 75.7	50.7 75.9	38.7 134	35.2 76.7	67.9 73.3	64.2 171	53.9 101	92.8 19.2
	SAINT-Early (Arxiv’25)	64.7 56.8	48.1 59.6	37.7 99.1	34.2 59.7	62.5 55.3	41.2 127	48.1 76.3	82.8 39.0
	QuietPrune (Ours)	67.0 55.1	50.4 56.2	39.1 94.6	40.3 56.7	69.0 52.8	77.3 119	57.2 72.4	98.5 42.1
Qwen3-VL-4B	No prune	81.7 60.2	63.1 72.8	50.6 159	55.1 258	84.1 83.9	85.2 362	70.0 166	- -
	FastV (ECCV’24)	72.1 56.3	48.9 65.8	47.2 137	45.9 191	71.4 75.4	38.9 284	54.1 135	77.3 18.7
	DivPrune (CVPR’25)	79.6 61.4	60.8 74.7	49.1 158	52.6 211	82.3 86.2	61.2 352	64.3 157	91.9 5.42
	PACT (CVPR’25)	80.3 57.2	61.2 62.9	48.8 133	52.4 173	82.0 71.2	70.2 265	65.8 127	94.0 23.4
	AIM (ICCV’25)	80.3 64.1	60.1 75.9	49.4 154	52.7 209	81.4 87.0	48.6 344	62.1 156	88.7 6.02
	VisPruner (ICCV’25)	75.6 55.7	54.2 64.9	44.2 133	45.3 174	71.5 70.9	22.5 256	52.2 126	74.6 24.1
	SAINT-Late (Arxiv’25)	80.0 54.8	59.2 63.3	48.0 129	51.3 179	80.8 73.8	69.0 268	64.7 128	92.4 22.9
	SAINT-Early (Arxiv’25)	73.1 49.1	52.9 62.7	45.2 114	43.3 150	70.7 65.8	36.7 223	53.7 111	76.7 33.1
	QuietPrune (Ours)	81.8 53.2	62.8 60.5	48.7 114	53.4 149	82.6 67.2	72.5 220	67.0 111	95.7 33.1
InternVL3-8B	No prune	85.2 206	68.6 210	56.9 441	48.2 203	85.4 179	88.2 535	72.1 296	- -
	FastV (ECCV’24)	82.5 127	62.9 134	54.1 250	45.6 131	81.1 120	81.6 298	68.0 177	94.3 40.2
	DivPrune (CVPR’25)	82.7 159	65.1 169	53.6 329	44.9 169	83.8 156	71.5 415	66.9 233	92.8 21.3
	PACT (CVPR’25)	83.6 129	65.2 131	56.1 244	48.3 133	83.7 122	82.8 292	70.0 175	97.1 40.9
	AIM (ICCV’25)	83.8 149	64.4 141	54.7 266	47.9 150	84.9 143	39.6 333	62.6 197	86.8 33.4
	VisPruner (ICCV’25)	78.2 126	56.5 131	53.7 238	37.9 132	74.7 121	36.9 287	56.3 173	78.1 41.6
	SAINT-Late (Arxiv’25)	84.0 134	63.8 136	55.1 250	48.1 136	83.6 125	79.9 304	69.1 181	95.8 38.9
	SAINT-Early (Arxiv’25)	81.3 113	60.2 117	54.2 211	42.7 116	79.6 106	41.2 253	59.9 153	83.1 48.3
	QuietPrune (Ours)	84.8 112	65.9 114	51.1 208	50.7 114	84.8 104	84.1 249	70.2 150	97.4 49.3

generalization on Qwen3-VL, particularly at high pruning rates. VisPruner’s unstructured pruning before the projector limits compatibility with VLMs using token merging operations. PACT, SAINT, and DivPrune maintain reasonable accuracy at high pruning rates but still underperform our method. These results demonstrate that our query-guided approach more effectively preserves query-aligned visual tokens, achieving superior accuracy.

Table 1 presents the detailed latency and accuracy results for all methods at a 50% pruning rate. On the InternVL3-1B model, QuietPrune maintains 98.5% accuracy and reduces 42.1%. In comparison, the best among other late pruning methods attains only 94.4% relative accuracy with 23.1% latency reduction. Although SAINT’s early pruning variant achieves comparable latency reduction, it suffers a significant accuracy drop. The performance gap narrows on larger models, but our method remains consistently superior across all evaluated architectures. Notably, on the OCRBench dataset, QuietPrune demonstrates the largest accu-

racy advantage. This is because OCRBench targets fine-grained textual regions that are easily discarded without query-aware guidance. Our query-guided strategy selectively preserves visual tokens containing such critical details, thereby improving overall model performance.

5.3. Ablation study

To verify the effectiveness of each component, we perform ablation studies on InternVL3-1B and Qwen3-VL-4B at a 50% pruning rate across six benchmarks. As shown in Tab. 2, we systematically evaluate: (1) semi-structured vs. unstructured pruning, (2) query-guided vs. visual-saliency-based token selection, (3) inverse transformation vs. random adapter initialization, and (4) the impact of redundant token aggregation. All results are reported as relative accuracy to un-pruned baselines.

Semi-structured pruning: As shown in Tab. 2, with semi-structured pruning, the models achieve 98.45% and 95.71% relative accuracy on InternVL3-1B and Qwen3-VL-4B, respectively. In contrast, unstructured pruning yields only

Table 2. Results of ablation study. “SS” refers to Semi-Structured versus unstructured pruning; “QG”, namely Query-Guided, indicates pruning based on textual-visual relevance versus visual saliency alone; “IT” denotes adapter is initialized via Inverse Transformation versus random initialization; “TA” indicates whether redundant Token Aggregation is applied. “RA%” represents the Relative Accuracy. The Average Score (AS) is evaluated with a 50% pruning rate across six benchmarks.

Models	SS	QG	IT	TA	AS	RA%
InternVL3-1B					58.08	
InternVL3-1B with pruning		✓	✓	✓	47.16	81.20
	✓		✓	✓	54.79	94.34
	✓	✓		✓	55.29	95.20
	✓	✓	✓		56.89	97.95
	✓	✓	✓	✓	57.18	98.45
Qwen3-VL-4B					69.97	
Qwen3-VL-4B with pruning		✓	✓	✓	54.83	78.36
	✓		✓	✓	62.81	89.77
	✓	✓		✓	64.91	92.77
	✓	✓	✓		66.38	94.87
	✓	✓	✓	✓	66.97	95.71

81.20% and 78.36% relative accuracy on the same models. This result confirms that unstructured pruning will distort patch positional information, and our method successfully preserves spatial consistency.

Query-guided pruning: As shown in Tab. 2, query-guided pruning improves relative accuracy by 4.11% and 5.94% over visual-saliency-based pruning on InternVL3-1B and Qwen3-VL-4B, respectively. Figure 3 shows that tokens retained by visual-saliency-based pruning tend to concentrate on foreground objects and dominant image regions. In contrast, pruning based on visual-textual relevance preserves tokens that align with the semantic content of the input query. Consequently, when the query refers to background elements or fine-grained details, our approach yields substantially more accurate predictions.

Inverse transformation: Using inverse transformation to initialize the adapter improves relative accuracy by 3.25% and 2.94% compared to random initialization. This demonstrates that inverse transformation positions the adapter in a more favorable optimization landscape, enabling faster convergence with minimal training on a small amount of data.

Redundant token aggregation: As shown in Tab. 2, redundant tokens aggregation brings accuracy improvements of 0.50% and 0.84% on InternVL3-1B and Qwen3-VL-4B, respectively. Although not substantial, aggregating redundant tokens has a negligible impact on latency. We incorporate this component into our method to achieve optimal performance.

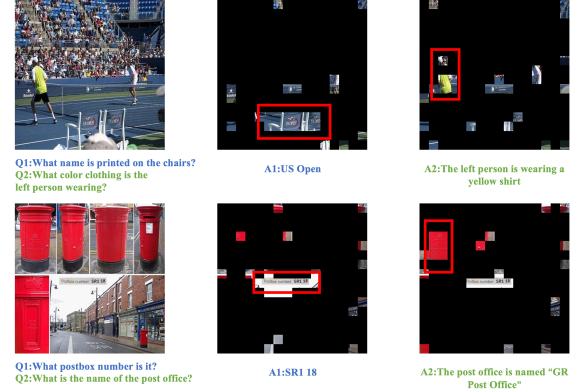


Figure 7. QuietPrune results on InternVL3-1B model. Our method adaptively retains the visual tokens that are relevant to each query.

5.4. Visualization of pruning results

To further validate the effectiveness of the proposed query-guided framework, we visualize the token pruning results of QuietPrune under a 90% pruning rate on InternVL3-1B model. As illustrated in Fig. 7, the red bounding boxes highlight the visual tokens retained that are relevant to each query. This result demonstrates that our adapter effectively maps query tokens into the visual domain and that our visual-textual relevance score accurately captures the semantic alignment between visual tokens and the given query. Consequently, even under high pruning ratios, our method maintains strong relative accuracy by preserving the most relevant visual content. Additional visualization examples are provided in the supplementary materials.

6. Conclusion

In this paper, we propose a novel query-guided early pruning framework to accelerate inference in VLMs by selectively discarding query-irrelevant visual tokens within the ViT. We design a text-to-vision adapter by leveraging an inverse transformation of the pretrained vision-to-text projector in the VLM to generate a [Q-CLS] token for ViT pruning. Furthermore, we utilize the attention score between the [Q-CLS] token and the visual tokens as a measure of text-visual relevance to perform semi-structured pruning and redundant token aggregation with negligible computational cost. Extensive experiments show that our proposed adapter effectively projects textual queries into the visual domain, enabling query-aware token selection. Consequently, our semi-structured pruning strategy adaptively retains visual tokens relevant to the input query while preserving spatial consistency, thereby maintaining high accuracy. On different benchmarks, our method consistently outperforms the existing methods in both relative accuracy and latency reduction across different VLM architectures and model sizes. Our work lays the foundation for more efficient early pruning strategies in VLMs.

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